

DISCUSSION

SESSION 1

SIR WILLIAM PENNEY, *Session Chairman*

I AM honoured by being asked to take the chair at the first session of this conference attended by distinguished technologists from many countries. The purpose of this session is to hear papers from four countries and the Euratom community. They discuss the broad plans for the development of the fast reactor system and the place that fast reactor power stations are expected to have in national electricity generating programmes.

I shall be paying careful attention to the views of speakers at this conference on two points: one—What degree of confidence can be justified today in the dual objectives of the fast reactor system to produce cheap electricity and to conserve uranium? Two—What measure of agreement is there today that the fast reactor system will not diversify because there are different choices of materials?

In the United Kingdom there is a strong consensus of opinion that the plutonium-uranium ceramic fuel and sodium cooling have a clear lead over other materials. We think that we shall be able to make faster progress by building a prototype based on these materials rather than by making a costly and prolonged effort on further research. Other countries, taking account of their own particular circumstances, may take a different view. I am sure that we shall all be interested to hear these different opinions, together with the reasons.

I now have great pleasure in calling on Mr. Searby to deliver the paper by himself and Mr. Iliffe.

Presentation of Papers

P. J. SEARBY—*Paper 1/1*

The main aim of this paper is not to indicate what the place of the fast reactor will be in the future U.K. Power Programme; it is partly to suggest what that place can be and it is also to draw attention to the importance of looking at the fast reactor in the context of the whole system of which it is a part. This applies both to its development and to its subsequent commercial exploitation and includes a consideration of its relationship not only to the characteristics and performance of alternative and complementary nuclear power stations but also to the size, and rate of growth, of the electricity system of which it will form a part.

The interaction of the fast reactor with the whole power system is extremely

complex, but to assist us in discussing the position, and in stressing the importance of considering the whole system, we have prepared a simplified model for analysing the situation. Clearly, the position will never be as uncomplicated and static as we have had to make it appear, but analysis along these lines is, we suggest, essential if the maximum advantage is to be taken of the fast reactor. Frequent reassessment of the position will be required as the background into which the fast reactor is to fit becomes clearer. The reactor designer is perhaps already used to the fact that from time to time the economic ground rules against which he assesses his reactor's development programme may change.

However, though our primary aim has been to suggest the importance of a whole system approach and to indicate a simplified form of such an approach, we have sought to use our analysis to produce what we believe to be a realistic study of how the future position in the U.K. might develop.

A. DE STORDEUR—*Paper 1/2*

I will limit the first part of my presentation to summarizing very briefly the elements of the paper which has been distributed on the review of the fast breeder development in the Community.

The efforts in this field in the Community are now entirely carried out in five associations between nuclear national organizations and Euratom. These associations are staffed and funded jointly by the national governments and Euratom, the participation of the governments being about two-thirds while that of Euratom is one-third, and the total rate of expenditure at the present time is roughly 50 million dollars a year. This figure gives you an idea of the importance of the research and development now going on in the Community in this field.

The overall objectives of the different programmes is to achieve competitive fast breeder reactor power by the late 1970's, with reactors of a size of 1000 MW(e).

Turning now to the strategy aspect of the fast reactor in the power economy and its consequences on the natural uranium demand, I want to present the elements of a recent study* which we carried out in Brussels in order to establish a comparison, on the basis of their natural resources demand, between thermal converters and uranium fuelled fast reactors as potential plutonium producers in the mixed nuclear economy which will exist before plutonium fast breeders alone can satisfy the power demand.

Under the assumptions used, this comparison has indicated that uranium fuelled fast reactors do not introduce a larger maximum cumulated demand

*J. VILAIN, Natural resources demand—enriched uranium startup of fast reactors, Document EUR/3080/66 e (to be published).

of natural resources than the best thermal converters, even though their specific inventory is much larger. As soon as they can be built economically, and assuming as a first approximation that a uranium-235 fuel cycle in a fast reactor would not differ appreciably from a plutonium fuel cycle cost, it would therefore seem acceptable to make good use of natural resources, to start fast breeders with enriched uranium. A complete analysis of the interchange ability of enriched uranium and plutonium cores in the same reactor and of the fuel cycle costs would be needed to check the validity of such a procedure. This is being made by the Karlsruhe and Belgian Euratom associations and, if the results are positive, the consequences of this procedure on the demand for enriched uranium and the availability of separation plants for some 20–30 years to come will have to be examined.

The basic assumptions used in the simplified analytical treatment of the mixed nuclear economy are as follows:

(a) The “demand function” $P(t)$ representing the future total installed nuclear capacity of the system as a function of time is a continuous rapidly increasing function; its analytical nature (exponential, power law, etc.) need not be specified in the calculations. Its instantaneous doubling time

$$\tau(t) = 0.69 \frac{P(t)}{\frac{dP(t)}{dt}}$$

is a continuous slowly increasing function which eventually reaches the spontaneous doubling time (τ_{lim}) of a system of plutonium breeders alone at a finite time (t_{lim}) (e.g. towards year 2010–2020). The time derivative $d\tau/dt$ is always small compared to unity, this being the order of magnitude of the approximations involved in the calculations.

(b) The demand function $P(t)$ is satisfied by a single type of converter reactor and a single type of plutonium breeder reactor of given invariant characteristics, the plutonium for the new breeder inventories being produced in both; this allows a first comparison between converters in an actual system where several types of converters will undoubtedly be used simultaneously, and where the characteristics of breeders will improve with time (doubling time decreasing as technology permits).

(c) Instantaneous plutonium balance is achieved at all times, new plutonium breeders being built to the full extent of the plutonium made available, with no stockpiling. Delays and losses between in-pile production and reuse are taken into account.

(d) These delays are fairly small compared to the doubling times, so that the differential equations obtained for the converter total capacity $T(t)$ and the plutonium breeder total capacity $F(t)$ can be made linear.

One derives approximate simple linear first order differential equations

with constant coefficients for $T(t)$ and $F(t)$; their second members are linear expressions in $P(t)$ and $dP(t)/dt$.

Before plutonium breeders take over completely (when $\tau(t) > \tau_{\text{lim}}$), $T(t)$ increases at first, goes through a maximum $T_m = T(t_m)$ and eventually decreases to zero; whereas $F(t)$ is always increasing.

On the basis of the assumption $d\tau/dt \ll 1$, one can derive simplified solutions $T(t)$ and $F(t)$ of the differential equations which are linear in $P(t)$, rational in $\tau(t)$ and involved no time integral over $P(t)$. They allow one to obtain the approximate results which follow:

(a) The maximum of $T(t)$ always occurs at approximately the same time

$$t_m = t_{\text{lim}} - \frac{\tau_{\text{lim}}}{0.69}$$

at given plutonium breeders characteristics and demand function, whatever the converter.

(b) When plutonium breeders are introduced early, the height T_m of the maximum varies roughly as the inverse of the specific plutonium production Q_1 (e.g. kg Pu/MW(e)/year) of the converters.

(c) When additional delays Δt_0 to the introduction of plutonium breeders occur, the height of the maximum is increased by an additional term which is exponential in Δt_0 . Comparatively small delays (e.g. $\Delta t_0 \approx 10$ years) can cause a significant increase of T_m , so that an early introduction of plutonium breeders seems imperative.

(d) The equivalent annual demand in natural uranium from the converters

$$\delta(t) = CT(t) + J \frac{dT(t)}{dt}$$

is the sum of a consumption term $CT(t)$ where C = equivalent specific consumption (e.g. ton U_{nat} /MW(e)/year) and of an incremental inventory term $J[dT(t)/dt]$ where J = equivalent specific inventory (e.g. ton U_{nat} /MW(e)). $\delta(t)$ increases at first, goes through a maximum at about the same time $t = t_m$ as $T(t)$, and drops sharply to zero afterwards.

(e) The height δ_{max} of the maximum of $\delta(t)$ is approximately equal to CT_m , and is independent of the inventory term:

- (i) when plutonium breeders are introduced early, it varies as C/Q_1 , i.e. as the inverse of the equivalent conversion ratio of the converters;
- (ii) when plutonium breeders are introduced late, it has an upper limit which varies roughly as $C/(Q_1 - Q_2)$ where Q_2 = specific plutonium production of the breeders (this at given breeder characteristics and demand function).

(f) The cumulated equivalent consumption $\mathcal{D}(t) = \int \delta(t) dt$ increases

sharply at first and levels off after t_m . The height of its maximum is comprised between limits which also vary as C/Q_1 or $C/Q_1 - Q_2$ at given plutonium breeder characteristics and demand function. Therefore, when $T(t)$ goes through a maximum, one obtains indexes of consumption of natural resources

$$\frac{C}{Q_1} \quad \text{and} \quad \frac{C}{Q_1 - Q_2}$$

for the comparison of converters over the full development span of a mixed economy.

In the initial phase, where converters are necessary, the index of comparison would be, approximately:

$$\frac{C + 0.69/[\tau(t)] J}{Q_1}$$

From the numerical values of all previous symbols for different reactor types, it can be shown that under the assumptions used, enriched uranium fast reactors (used as converters) have as good indexes of consumption of natural resources as the best thermal converters (heavy water natural uranium type).

The calculation would have to be refined beyond the reach of the analytical procedure employed here to take into account decreasing doubling time of plutonium breeders during the period of the mixed economy. It is thought that this improvement of the quality of fast reactors could only reinforce the conclusions drawn here from the simplifying assumption of constant breeder characteristics.

G. VENDRYES—*Paper 1/3**

I should like to indicate the French attitude to the questions raised by Sir William Penney.

To what extent can one rely on the fast neutron reactor to produce electricity cheaper than any other type of reactor? In 1966 one can only express a very cautious judgement considering that it is not for some 10–15 years that we expect a final economic demonstration. We must, however, stress that from the point of view of investment and that of fuel cycle, fast reactors have advantages which should enable them to be competitive with the best thermal reactors and as these thermal reactors have a lower fuel cost than conventional stations one can accept that fast reactors, while combining the best advantages of the thermal reactors, will be the most economic of all. We have worked out economic calculations, but these are open to doubt since they are based on long term perspectives.

* Originally appeared as Paper 1/8.

An important point that I, for one, doubt, is that the fuel cycle has an economic future in fast reactors. There are certain unknown factors concerning not only the rate of burnup that can be achieved in larger stations but also the effect of the integrated flux on structural material. We have insufficient information on this at present and it will take some time to acquire.

Sir William's second question referred to the technical diversity of the various models of fast reactors, particularly from the point of view of materials. Here our position is quite similar—I would not say identical—to the British position.

As far as fuel is concerned, we think that stress should be laid on plutonium. It is the obvious choice of fuel for fast reactors and that is why we have selected it from the outset for Rapsodie. I consider the possibility of reverting to U-235 is far from being an urgent question. We must first demonstrate that plutonium fuelled fast reactors are competitive and economic and this should be done within the next 10–15 years. Only when it appears that the fast reactors are so superior to the thermal reactors that it is better to use U-235 in them than in the thermal reactors—then, and only then, will the question arise. If it does arise, it will lead to a higher consumption of enriched uranium which would run into tens of tons and would create an important problem of supplies. But this is a question which will not arise for some 10 years.

I shall not take up the point about the form in which plutonium will be employed as I have already said something about it in the paper. We have already chosen a mixed oxide, $\text{UO}_2\text{-PuO}_2$ for Rapsodie, which we have already studied, but we are also looking into the question of carbides which we consider as a possible alternative solution. We have chosen sodium as the heat transfer fluid and shall retain it for the reasons I have just given.

The last point raised by Sir William Penney—Is it better, in order to progress in this field of fast reactors, to build a new prototype or should we spend a number of years in intensive research work? We feel that the point has been reached where sufficient technical development has been achieved to envisage in the forthcoming years—and as far as we are concerned the beginning of 1969—the construction of a prototype of significant power. We think this is the best way, if not the only way, of handling the technical problems which have to be solved for the development and success of the series. However, we recognize the value of a parallel programme on research and development.

M. SHAW—*Paper 1/4*

The primary objective of the civilian power reactor development programme in the United States is to achieve widespread commercial use of nuclear energy for the production of heat and electricity, while fully exploiting the

energy available in our resources of uranium and thorium. The Commission's* objective also includes fostering the development of a self-sufficient and competitive nuclear industry.

The successful achievement of the primary objective will require the development of fast breeder reactors and their successful introduction into commercial use. To accomplish this, one of the highest priorities has been given by the Commission to planning and executing the research and development programme required for mastering the technically difficult and challenging sodium cooled fast breeder reactor concept in a timely manner. This priority was determined, after an assessment was made of the status of various breeder concepts and of the overall liquid metal, fast breeder reactor programme, and each of its programme elements. On a lesser priority basis, work is continuing on other coolant concepts.

Based on this assessment, formal intensive and comprehensive reviews are being made, in each technical area of the LMFBFR programme, to determine the major needs of the programme, its deficiencies, and the plans required to eliminate these deficiencies and further strengthen the programme. The initial reviews have been completed on fuels and materials, physics and their associated facilities. Reviews are continuing in the areas of reprocessing and recycling fuels, safety, components, instrumentation and control, sodium technology, fuel handling, plant design and, in each case, the related facilities.

Y. IHARA†—*Paper 1/5*

Because of the lack of the major energy resources, Japan has had to depend on imports of energy, and the ratio of dependency on the overseas supply has more than doubled in 5 years. The principal imported energy source is oil, the total amount of which will reach about 600 million kilolitres by 1985 should Japan fail to utilize nuclear power on a large scale by that time.

In establishing the energy policy, the principles of both cheapness and security of supply must be considered, and nuclear power generation can possibly embody these two principles. Further, compared with oil fuelled power generation, nuclear power generation does not need so much foreign currency.

R. A. MUELLER—*Paper 1/6*

At the end of 1964 the Karlsruhe group completed the first design study of a 1000 MW(e) sodium cooled, fast breeder reactor called Na 1. This design was used as the basis of a subsequent system analysis which showed the

* United States Atomic Energy Commission.

† The author, H. Murata, was not able to attend the conference.

dynamic behaviour, the safety features, and the effects on costs and safety of a number of design variations. Now a second design study, Na 2, is under way which will give the basic specifications for the prototype reactor of 300 MW(e).

In the same general pattern, a design study of 1000 MW(e) steam cooled, fast breeder reactor, D 1, has now been completed. Some of the results of this design study are in this paper. According to our principle, we do not claim this design to be the optimum of all possible steam cooled, fast reactors. We thought it more valuable to study one reference design first in more detail with the purpose of finding the physics and engineering problems typical of steam cooling. The more detailed optimization will again be made during the system analysis in the paper.

R. V. MOORE—*Paper 1/7*

From the papers already presented in the first session, it is clear that fast breeder power stations in all probability will be built in rapidly increasing numbers during the last two decades of this century.

In the United Kingdom, our work in the field of fast reactor development, particularly our experience over the last 4 or 5 years, leaves us in little doubt that the sodium cooled, fast reactor is technically feasible and can be engineered into large power units. Furthermore, as shown in Paper 1/1, a programme of such reactors in the United Kingdom should achieve substantial savings when compared with complete reliance on fossil fuelled, or thermal reactor stations.

The decision earlier this year to construct the 250 MW prototype at Dounreay commits the United Kingdom firmly to sodium cooling, at least for all the early fast reactor power stations. It is perhaps reassuring, for it is a big decision, to find that we do not venture down this road unaccompanied.

The U.S.S.R. is already constructing her prototype, the sodium cooled BN 350. Indeed there seems to be a remarkable uniformity of view on the parameters and general design for a reactor of this type.

However, industrial exploitation will not start in a big way for a further 5 or 10 years, but then it will probably continue well into the next century, when fast reactors are likely to be the main type of reactor to be built.

In spite of the firm position of sodium cooled, fast reactors now, on this long time scale, it would be perhaps unwise to assume that they will continue exclusively to occupy the prime place indefinitely, unrivalled by other possible fast reactor systems.

Consequently, while devoting the major part of our effort to the vigorous development of the sodium cooled, fast reactor, like other countries, the United Kingdom is carrying out a broader based research programme, studying other possible forms of fast reactor.

Discussion

C. A. RENNIE (OECD High Temperature Reactor Project): The papers at this conference have emphasized that breeding is a means to an end and not an end in itself—not in reactors, anyway. If breeding leads to lower fuel cycle costs and hence to lower generating costs than any other type of reactor, it will automatically lead to better utilization of natural uranium. If a fast reactor did not achieve lower power costs it would not be worth building, simply to conserve uranium. It would be cheaper to look for more uranium and cheaper methods of extracting it. Assuming that fast reactors will be competitive, one important question is how to make plutonium for them. This has been referred to by every speaker.

Table D1 shows the burnup, the rating and utilization of uranium which can be achieved with different thermal converter reactors. The burnup and utilization are given in megawatt years of electricity per ton of natural uranium which has to be mined and the rating in megawatts of electricity per ton of natural uranium which has to be mined. Expressed in these units the figures take account of the different fuel enrichments and different thermal efficiencies in these reactor systems, and the reactors are listed in order of increasing uranium utilization. In practice, the figures given for the HTR system are actually for a thorium cycle, but the figures for the slightly enriched uranium/plutonium HTR would be a burnup of 7.5 MW(e)y/tonne a rating of 5 MW(th)/tonne and a utilization of 7 MW(e)y/tonne. Thus the enriched uranium HTR would appear in the table between the Candu system and the thorium HTR system.

The megawatt years of electricity that can be produced per tonne of uranium mined increase from about 2.4 in a magnox reactor to 7 in an HTR plutonium producer. Plutonium production can be expressed in Kilograms per megawatt year of electricity. These figures are not shown but of course they are in reverse order; the magnox reactor has the highest production of plutonium per megawatt year of electricity, about 0.4 kg, and the HTR the lowest, about 0.12 kg. This is not surprising since reactors with good uranium utilization obtain this by having a good neutron economy and by burning some of the plutonium they produce in the reactor itself. If this uranium utilization is multiplied by the plutonium production, the answer will be the amount of plutonium produced for each ton of uranium mined. The result is that all these reactors give about the same production of plutonium, about 0.95 kg of plutonium per ton of mined uranium, within, say a plus or minus 10–15%. A rough working rule is that about 1 kg of plutonium will be produced for each ton of uranium mined.

At first sight it seems that it does not matter which type of reactor is used, but there is a great advantage in using a reactor with a good uranium utilization, as much more electricity is produced while making the same amount of

TABLE D1

Reactor type	Size MW(e)	Fuel enrichment	Feed factor	Thermal efficiency	Average rating MWd(th)/tonne	Average burnup MWd(th)/tonne	Burnup MWy(e)/tonne	Rating MW(e)/tonne	Utilization MWy(e)/tonne
Magnox	(1) 640	Natural	1.0	0.335	2.85	3,000	2.76	0.96	2.4
PWR	(1) 1000	3%	6.0	0.31	27.8	24,000	3.42	1.45	3.1
BWR	(1) 600	2.45%	4.8	0.315	15.1	22,000	3.94	0.99	3.3
AGR	(2) 1200	{ 1.6% initial 2.8% feed	3.0	0.415	9.5	19,000	5.15	1.32	4.3
SGHW	(3) 500	1.63%	3.0	0.32	12.8	18,000	5.25	1.36	4.4
ORGEL	(4) 250	Natural	1.0	0.31	10.9	6,300	5.33	3.35	4.9
CANDU	(5) Not stated	Natural	1.0	0.29	17	9,000	7.17	4.94	6.7
HTR	(6) 500	93%	201.0	0.43	{ 2.400 (2.6 MW(th)/kg)	1,730,000 (1.9 fifa)	9.84	5.15	9.0
HTR with recycle	(6) 500	93%	201.0	0.43	{ 2.100 (2.3 MW(th)/kg)	2,700,000 (3.1 fifa)	15.9	4.57	13.5

- (1) C. PLEINEVAUX, G. VAN BOCKSTAEL, Principales Caractéristiques Techniques des Centrales Nucléaires des Types Graphite-Gaz et Eau Légère Bouillante ou Pressurisée. EUR/2448/f 1965.
- (2) *Dungeness "B"* AGR Nuclear Power Station. Atom. No. 107/September, 1965. Monthly Information Bulletin of UKAEA.
- (3) N. L. FRANKLIN *et al.* Fuel Cycle Costs for a Large Nuclear Power Programme. Fuel Cycle Costs for Magnox, AGR and SGHW Reactors. *Third International Conference on Peaceful Uses of Atomic Energy, Geneva, 1964*, Paper A/Conf. 28/P/159a.
- (4) A. BAHBOUT, S. ORLOWSKI, Référence Design d'une Centrale Orgel de 250 MW(e). EUR/2190/f 1964.
- (5) *Power Reactor Technology*, Vol. 6, No. 4, Fall 1963.
- (6) E. SCHRÖDER, J. H. BLOMSTRAND, H. BRUNEDER, Fuel Cycle Costs for (HTGC) Power Reactors. Paper 14 presented at this Symposium.
- (7) The feed factor is defined as the number of tonnes of natural uranium required to produce one tonne of enriched uranium, assuming a reject concentration of 0.253% as in the *US AEC Price Schedule* published in 1962.

$$\text{Feed factor} = \frac{10 \text{ (Enrichment in \%)} - 2.53}{4.59}$$

- (8) The average rating is given in thermal megawatts per tonne of natural or enriched uranium fuel. For the HTR, the rating is also given in thermal megawatts per kilogram of fissile material.
- (9) The average burnup is given in thermal megawatt days per tonne of natural or enriched uranium fuel. For the HTR, the burnup is also given in fissions per initial fissile atom (fifa).

(10) The uranium utilization factor is defined here by:

$$\frac{1}{U} = \frac{1}{B} + \frac{0.05}{R} \quad \text{or} \quad U = \frac{B}{1 + \frac{0.05B}{R}}$$

plutonium for each ton of uranium. This is, of course, providing that the reactors are economic power producers. Here there is perhaps a problem because the present predicted power generation costs for the High Temperature Reactor as a plutonium producer are between 3.5 and 4 mills/kWh, of which about 1.2 mills/kWh is the fuel cycle cost. This is the same, approximately, as the power generation costs permitted for fast reactors in Papers 3/5 and 3/6.

The question is whether fast reactors are needed at all if the same power generation costs and a good uranium utilization can be obtained elsewhere. There is really no doubt about this. A fast reactor, even if it only has the same power generation costs as any other advanced thermal reactor will always give you a higher uranium utilization, so there is certainly a case for them. These figures of power costs both for the HTR's and fast reactors are predictions, as neither has been built on a large scale yet. The point is that one must give serious consideration when making plutonium for use in fast reactors to using a convertor reactor which has a good uranium utilization, as this, after all is the first step to conserving uranium resources. Also, a reactor with a good uranium utilization is far less susceptible to changes in the price of uranium.

However, it might again seem an advantage to use a reactor with a high rating as the plutonium producer because eventually the thermal reactors which produce plutonium will have to work on lower load factors than the fast reactors for which they supply plutonium.

Again this should be an HTR reactor for two other reasons. Firstly, if and when there is a surplus of plutonium, it could be burned quite efficiently in an HTR reactor. Secondly, if it turned out that fast reactors are not as economic as is hoped, HTR reactors can always be operated on the highly enriched uranium/thorium/uranium 233 cycle, which, in any case, gives low fuel cycle costs.

I have put forward these figures for the HTR without giving a justification for them, but they can be checked in the Proceedings of the Drajon Symposium on High Temperature Reactors.*

W. FRANKOWSKI (Institute of Nuclear Research, Warsaw): I would like to make a short comment on Paper 1/2 by Mr. de Stordeur. The analysis of the fuel consumption of a system composed of thermal convertors and fast breeders, when some simplifying assumptions are made, results in a first order differential equation mathematical model. This equation can be easily solved, provided the installed power is assumed to grow according to some simple, say exponential, function and that a constant load factor is assumed during the period of time investigated. These two assumptions are rather different from what can actually happen. But there is another simplification

* *J. Brit. Nucl. Energ. Soc.* (1966) July.

which the author discussed which can hardly be accepted; the assumption of constant doubling time equal to the equilibrium doubling time. This constant doubling time can be of the same order as the time necessary for a reactor to reach equilibrium. That is why the change of isotopes in the whole system should be taken into account.

In this case, we have to use a more sophisticated mathematical method using an integral differential non-linear equation. We have found that this equation can be solved for arbitrary shapes of the functions denoting the increase of installed capacity and the changes in the load factor. The results show that the plutonium production per unit of electrical energy, for the particular convertor reactor investigated, is crucial as far as the total fuel need of such a combined system is concerned. Substantial shifts can also be observed in the time necessary for such a combined system to become self-sufficient in its need for plutonium. Nevertheless, I think that a simple model is a useful first approximation.

H. M. CARRUTHERS (Central Electricity Generating Board, United Kingdom): In Paper 1/1, having developed this most useful method of assessment, the authors depend essentially on the accuracy of the data assumed, including the generating cost differential between the fast reactor and the thermal reactors. I would like to refer to an estimate of this differential made within the Board. Because of the difficulty of estimating absolute costs for power stations of the late 1970's and beyond, we thought it preferable to try to identify differences in capital cost between developed advanced gas cooled reactors and their contemporaries in the sodium cooled fast reactor field. We concluded that the effective capital cost of fast reactors would be £4/kW higher than the AGR's when the off-load refuelling of the fast reactors had been taken into account.

On the other hand, assuming zero value plutonium, the indications were that the fast reactor would have a fuel cycle cost advantage equivalent to some £15/kW. The net advantage of the fast reactor thus appeared to be about £11/kW, or 0.46 mills or thereabouts per kWh. This suggests that the 0.4 mills/kWh saving, quoted in Appendix 2 of the paper, may be in the centre of the uncertainty band rather than the minimum saving.

Nevertheless, 0.4 mills/kWh would amount to large sums by the end of the century and, of course, the beneficial effects of a reduced uranium consumption must not be forgotten. I would now like to query three points in Paper 1/1.

- (a) the proportion of installed capacity which could work at maximum load factor is shown in Table 2 as rising to 40% by 1990. This seems to imply a speculation that the Generating Board's system load factor will have improved significantly over the current figures;

- (b) the load factor order of merit is stated as magnox, fast reactor, and advanced thermal reactor, whereas, using the usual criterion of incremental running costs, the fast reactor would take preference over both thermal reactors;
- (c) the notional programmes shown in Figs. 2 and 3 involve commitments to build 12,000 and 16,000 MW of fast reactor capacity before the first commercial station has run for a significant time.

I suggest here that these programmes do not take account of the natural caution of a utility with a new reactor system. Either of these programmes would imply that by 1978 the utilities would have complete confidence not only in the nominal economics of fast reactors but also in their high availability and their freedom of siting on safety grounds.

Will the authors say whether any sections in the paper are particularly sensitive to these assumptions of load factor, order of merit, and rate of installation?

E. C. W. PERRYMAN (Chalk River): In Paper 1/1, the author, when talking about the differential in generating costs between thermal reactors and fast reactors, assumes that all such reactors will have almost the same capital cost. If this is correct, and I am inclined to think it is, then the fast reactor is going to exist purely and simply on the basis of its fuel cycle cost. The fuel cycle costs we have seen so far are of the order of 0.8 mills/kWh; I suggest that we are already achieving fuel cycle costs of this order today.

M. M. TAUBE (Institute of Nuclear Research, Warsaw): In Paper 1/7 the author speaks about new type reactors and especially a fast reactor which proposes liquid salt fuels using uranium and plutonium chlorides and where the direct coolant would be liquid lead. In our laboratory we have been working for several years on such a reactor but have recently found a new possibility. We consider that a new system is possible which uses liquid salt fuels (fused chlorides) and where the coolant is one of the components of the salts which boils during reactor operation, in other words, a "salt boiling reactor". There would be two possibilities in this case:

- (a) a really homogeneous system where the boiling coolant would be aluminium trichloride;
- (b) another system, quasi-homogeneous, would have liquid metallic mercury as coolant which would also boil during the operation of the reactor.

In both cases the pressure would be between 20 and 45 atm at a temperature of 700°C. Our opinion is only provisional as yet, but we think these reactors will be stable and will have specific advantages, namely that during a prompt rise in temperature (following a rise in power), part of the liquid fuel would

be removed from the active core—in the homogeneous system owing to the increase of bubbles of aluminium trichloride vapours—in the non-homogeneous system, of course, owing to mercury vapour bubbles.

P. J. MARIEN (OECD High Temperature Reactor Project): In referring to the note issued on "Definitions of Fast Reactor Breeding and Inventory" (Appendix I), my first comments concern the choice of breeding and inventory parameters which are most appropriate when fuel requirement problems are studied. In an article, "A Simple Method for the Comparison of the Utilization of Fuel Resources" (*Nucl. Engng* (1966) July), I have adopted the following rating and burnup parameters which can be used both for converter reactors and breeders using plutonium as the main fuel source:

- R* the initial fuel cycle rating is the electrical power which can be installed per kg plutonium inventory;
- B* the burnup, or "breedup", is the electrical energy produced, expressed in MW(e)/net kg of plutonium burned in converter reactors or gained in breeder reactors.

I have found these parameters extremely useful in comparing the plutonium utilization in different converter and breeder reactor types on utilization plots giving B/β as a function of R/α , where β is the load factor and α the growth rate of the reactor types which are represented.

The parameter *R* has been included in Appendix I and is called the Inventory Rating. I would like to add to the parameters, already adopted to describe the breeding in fast reactors, namely breeding gain and breeding ratio, a third which I shall refer to as "breedup" in analogy to burnup, and which would give the ratio of the amount of electrical energy produced to the weight of plutonium gained; *B* is best expressed in MW(e)/kg plutonium. This parameter could be extremely useful in assessing the plutonium requirements of different power programmes.

There are two definitions given concerning doubling times. When one is interested to know how fast the power installed in fast breeder reactors can be doubled without external plutonium supply, the linear doubling time appears to be rather pessimistic whilst the exponential doubling time is optimistic. What we propose is a simple modification to the definition of the exponential doubling time to obtain a more realistic value, taking account of the possibility of reinvestment of plutonium fuel in new reactors after a certain time delay. Introducing θ as the average time between the production of plutonium in fast breeders and the reinvestment of the bred fuel in new reactor charges, the "delayed exponential doubling time" is given by

$$t_{DD} = 0.693 \times \frac{B/\beta}{R} \times 2^{\theta/t_D}$$

Figure D1 shows how the linear, the exponential, and the delayed exponential doubling times compare for different values of $B/\beta R$ and θ .

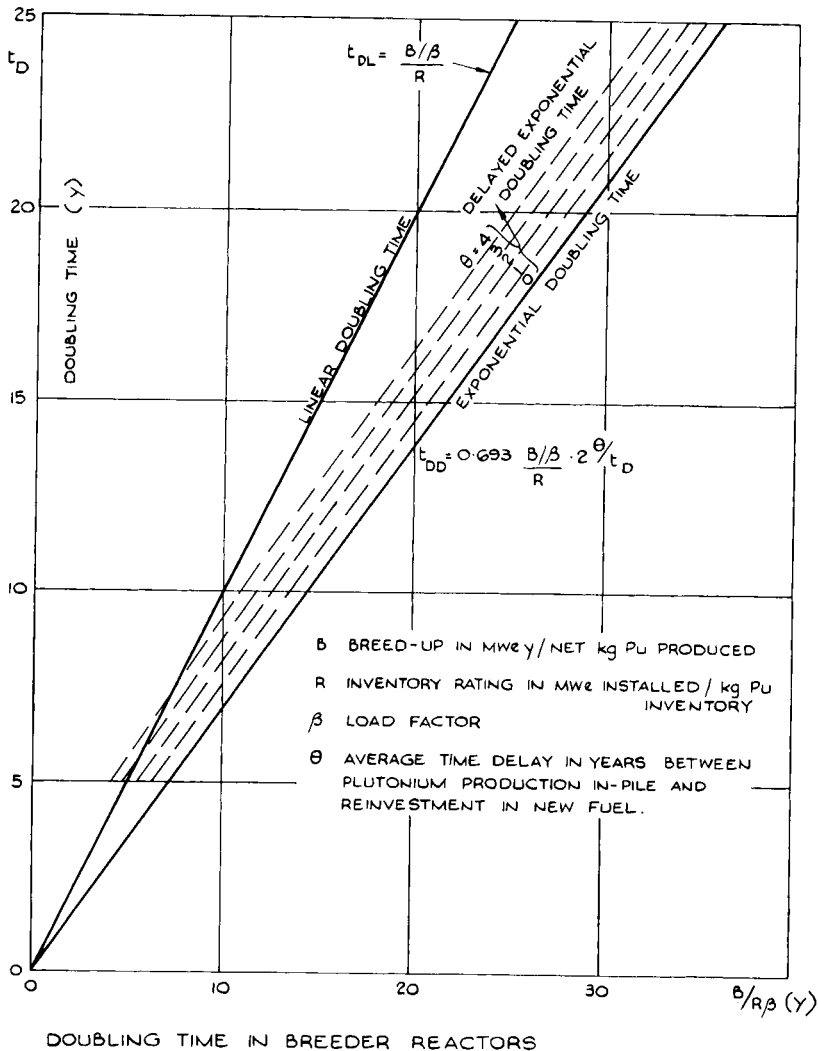


FIG. D1

In a practical case based on the data published in section 7 of Paper 1/1 we obtain the following characteristic parameters and natural doubling times:

R	inventory rating	0.25 MW(e)/kg Pu
B	breedup	2.74 MW(e) y/kg Pu
β	load factor	0.8
θ	delay time	1.5 years (assuming 2 year in-pile time and 6 months for reprocessing and refabrication)
t_{DL}	linear doubling time	13.7 years
t_{DE}	exponential doubling time	9.5 years
t_{DD}	delayed exponential doubling time	10.5 years

Authors' Replies

C. E. ILIFFE—*Paper 1/1*: In reply to Mr. Rennie, it is agreed that on his present estimates the HTR could well be in a good competitive position as a possible alternative to the fast reactor. The HTR, however, would suffer from any increase in ore prices, albeit less than most other types of thermal reactor. This possibility might well offset any small advantage the HTR might prove to have over the fast reactor. Mr. Rennie also suggested the use of an HTR as a substitute for plutonium producing thermal reactors, and in this connexion he quoted a plutonium production rate of $0.12 \text{ kg} = 239/\text{MW(e)}$ year. This figure is lower than that of any currently proposed thermal reactor known to the authors and it would need to be greatly improved if the objective of the system was to produce large quantities of plutonium in a fairly short time.

While the authors note Mr. Carruthers's view that the savings achieved by the introduction of fast reactors might well be nearer their lower estimate, they nevertheless consider that taking account of overall trends, their higher estimate is still equally likely. However, they are pleased to note that he considered even the smaller saving well worth while.

Mr. Carruthers also expressed the view that base load capacity might increase rather more slowly than was assumed in the paper. In fact, a smaller increase in base load would not significantly affect the average load factor of fast reactors because they do not reach even the current figure of 25% of capacity on base load until approaching 1990. Therefore, a smaller base load would only be significant in the later years of the programme, when the effect would in any case be reduced by the discounting techniques employed. However, a slower increase in base load capacity would certainly reduce the average load factor of the advanced thermal reactors and in consequence there would be some decrease in thermal reactor plutonium production. On the other side, the benefits obtained from plutonium recycling in thermal reactors would also be reduced and without repeating the entire calculation it is not possible to determine the relative effects on fast and thermal reactors, although it seems likely that the advantage of fast reactors would remain more or less unchanged.

In reply to Mr. Carruthers's query as to the position of the magnox station in the load factor order of merit, the simplified analysis of the Appendix required that the fast and advanced thermal reactors were next to one another in the order of merit, thereby ensuring that any exchange of capacity between the two resulted in a corresponding exchange in electricity generated. In consequence, the magnox station either had to be placed at the top or at the bottom of the order of merit. The calculations were carried out for both alternatives and the difference in their results was found to be negligible; the magnox programme represents only a very small part of the ultimate nuclear programmes considered. Although Mr. Carruthers expressed some concern that the initial rate assumed for the introduction of fast reactors was somewhat rapid, the authors feel that the benefits now being claimed for fast reactors if achieved would justify this rate and perhaps indeed an even faster rate, particularly in a situation of increasing plutonium stocks.

Mr. Perryman appeared to accept the assumption of Appendix II that fast and thermal reactors would have equal capital costs and then went on to point out that there was a thermal reactor system today which could give fuel costs as low as those of the fast reactor. Whilst it is true that Appendix II does make the assumption of equal capital costs, it also goes on to say: "The lowest uranium cost will be achieved by thermal reactors giving the greatest neutron economy. Such reactors may never, in practice, be installed to any appreciable extent since they generally suffer overriding disadvantages in either capital cost or burnup, or both." In the absence of further information from Mr. Perryman the authors are unable to exclude the possibility that the reactor system he has in mind may be one of these.

J. VILAIN—*Paper 1/2*—replied on behalf of A. DE STORDEUR: The formulation of our model was deliberately kept simple. Because of this it allows an approximate but fairly general comparison between possible converter types in a mixed economy, where converters and breeders coexist in dynamic equilibrium, and on the basis of the annual or integrated demand for natural uranium by the system.

There is, as a rule, an intrinsic difficulty in setting up any model:

- (a) one either assumes that the time variation of the installed nuclear capacity can be expressed by a simple, but arbitrary, function, for instance an exponential or a power law; or
- (b) one defines the "demand function" numerically, by means of some curve, or by assigning definite values to the installed nuclear capacity at certain future dates.

The problem can then be solved, sometimes algebraically, more often numerically, on the computer. In the latter case, the model can be made quite elaborate, if one wishes, and perhaps more realistic. However, strictly

speaking, the results obtained either way are valid only for that particular demand function which one adopted in the beginning. Presumably, one would have to compare a number of particular cases to be able to draw general conclusions which would not depend on the nature of the demand function.

We approached the problem in the opposite manner. We left the demand function $P(t)$ unspecified and were only able to do so because the model and its formulation were simplified. The only significant assumption, regarding the demand function, is that its instantaneous doubling time $\tau(t)$ is a slowly increasing function; in other words, the rate of expansion of nuclear power decreases regularly. From the model we derive simple linear first-order differential equations for the functions $\tau(r)$ and $T(t)$ which describe the time variation of the installed breeder and converter capacities. These equations result directly from the assumption of instantaneous plutonium balance, taking delays and losses into account. Their solutions can be approximated by simple expressions, linear in $P(t)$ and rational in $\tau(t)$. So can the expressions of the equivalent annual demand and equivalent accumulated demand for natural uranium by the system. As a result the latter quantities have in fact the "indexes of consumption"

$$\frac{C}{Q_1} \quad \text{or} \quad \frac{C}{Q_1 - Q_2}$$

to which they are proportional, at given demand function and breeder characteristics, and which then depend on the converter characteristics only. Those indexes permit a comparison to be made of the quality of the converter types in a mixed economy, in terms of quantity of natural uranium to be supplied to the system. The comparison is quite approximate, but independent of the exact nature of the demand function.

In answer now to Mr. Marien's previous comments, one can say that our model automatically takes his corrections into account. The reactors are defined in terms of specific plutonium production per MW(e)y of specific inventory and specific consumption, either of plutonium or of equivalent natural uranium.

Delays and losses are taken into account in the formulation of the model. Then, for instance, the breeding ratio that one uses in the calculations, or the doubling time, result from simple mass balance considerations and would, by definition, be those effectively obtainable from the reactors when in operation.

Written Contribution

J. J. WENT (Naamloze Vennootschap tot Keuring van Electrotechnische Materiaien, The Netherlands): In Paper 1/1, section 2, it is stated that nuclear

power is the cheapest source of electricity. I think in principle it is necessary to be very careful in making such assertions. Apart from the difficulty in calculating the exact cost of electricity from different sources, because of power load availability, etc., it is most probable that the price of oil or gas used to produce electricity will be adjusted in line with the price of other energy sources. Again, taking into account that the tax on oil puts it at a disadvantage, it is very probable that the price of natural gas will be adjusted to the price of oil. With nuclear energy the production cost is estimated without taxes.

In section 3 of the same paper, it is stated that the fast reactor will become the predominant reactor system in due course. This prediction is related to the fact that fast reactors will have much lower fuel cycle costs than any thermal reactor can be expected to achieve. For the fuel cycle costs, not only are the raw material costs important, but also the fuel element fabrication and reprocessing costs. If thermal reactors could be constructed with liquid fuels, it is probable that fabrication and reprocessing would be very much lower than for fast reactors and that they could be preferred in this respect. Secondly, in general the fissile material investment per MW(e) installed capacity is often much lower in thermal reactors than in fast reactors. Therefore, in principle, lower fuel cycle costs can be expected in liquid fuel reactors.

Author's reply

C. E. ILIFFE: Section 2, Paper 1/1 actually states that nuclear power will be the cheapest source of base load electricity after 1970. Thus full account was taken of Professor Went's point about load availability; the current assessments in this country, the U.S.A., and West European countries make it clear that the statement made in our paper is now widely regarded as correct. The slight advantage on generating costs sometimes quoted for conventional stations using oil without tax depends entirely on the use of very cautious assumptions about the life and load factor of nuclear stations, and the use of relatively high rates of interest. Thus, while we agree that it is necessary "to be very careful" with such statements, given the current trends it would, in our view, be wrong to over-qualify our statement.

On the queries on section 3, even if liquid fuelled thermal reactors should turn out to have lower fabrication and reprocessing costs per unit sent out than fast reactors, the fissile material cost is nevertheless the most significant item. (It is of the order of 70% of total fuel cycle cost in enriched thermal reactors.) Although the closed system type of analysis described in our paper makes it unnecessary to ascribe a value to plutonium, it does take into account the relative fissile requirements of fast and thermal reactors. The saving in system generating cost quoted in the paper and in Appendix II was derived by comparing fast reactors with plutonium burning thermal reactors, on the assumption that the plutonium produced within the system was fully utilized

in both cases. It follows that, as regards inventory, the analysis takes account of fissile material investment in fast reactors by limiting the number of fast reactors installed to the amount of plutonium available; thus, higher fissile material investment in fast reactors affects the electricity generated by the fast reactor installation as a fraction of total nuclear installation.